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How Do I Modify SAS®9 Programs to Run in SAS® Viya®?

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Abstract

SAS® Viya® is a modern, cloud-enabled analytics platform built for high-speed, scalable processing- but what does that mean in practice for SAS® programmers? This demonstration demystifies SAS® Viya® by explaining its core architecture, introducing caslibs, and showing how they compare to traditional SAS® libraries. You'll also see how to load data into memory and explore key considerations when working with the CAS server. Whether you're new to SAS® Viya® or transitioning from SAS® 9, this session provides a practical foundation for working effectively in SAS® Viya® .

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ORDERS Data Dictionary

You will work with a dataset containing customer information for specific orders from the fictitious company Orion Star.

Alphabetic List of Variables and Attributes in ORDERS

#	Variable	Type	Len	Format	Label
9	City	Char	180		City Name
13	Continent	Char	180		Continent Name
21	Cost	Num	8	DOLLAR13.2	Cost
12	Country	Char	120		Customer Country
4	Customer_BirthDate	Num	8	DATE9.	Customer Birth Date
5	Customer_Group	Char	240		Customer Group Name
2	Customer_ID	Num	8	12.	Customer ID
3	Customer_Name	Char	240		Customer Name
6	Customer_Type	Char	240		Customer Type Name
17	Delivery_Date	Num	8	DATE9.	Date Order was Delivered
22	Discount	Num	8	PERCENT.	Discount in percent of Normal Total Retail Price
23	Employee_ID	Num	8	12.	Employee ID
24	OrderType	Num	8		
16	Order_Date	Num	8	DATE9.	Date Order was placed by Customer
1	Order_ID	Num	8	12.	Order ID
10	Postal_Code	Char	60		Postal code
18	Product_ID	Num	8	12.	Product ID
19	Quantity	Num	3	COMMA12.	Quantity Ordered

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#	Variable	Type	Len	Format	Label
20	RetailPrice	Num	8	DOLLAR13.2	Retail Price
11	State_Province	Char	152		State Name
7	Street_ID	Num	8	12.	Street ID
8	Street_Name	Char	272		Name of Street
14	xyContinentLat	Num	8	15.6	xyContinentLat
15	xyContinentLon	Num	8	15.6	xyContinentLon

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Exploring SAS Studio

1. Launch Google Chrome from the desktop and select **SAS Studio** from the Bookmarks bar.
2. Enter **student** in the **User ID** field and **Metadata0** in the **Password** field. Click **Sign in**. Select **Yes** when prompted to opt in to all assumable groups.
3. Click the **applications menu** icon in the upper left corner. The applications menu shows all the available applications. Depending on your environment, these applications might differ. Select **Develop Code and Flows** to open SAS Studio. We use SAS Studio to write and submit code in this workshop.
4. The Start Page tab is open by default, and it provides convenient links to start writing a program or use other point-and-click tools, including importing and querying data. Click **Program in SAS** to open a new tab with the Program Editor. Each item that you open in SAS Studio has its own tab.
5. The navigation pane is on the left side of SAS Studio. Each section in the navigation pane provides access to different resources in your SAS environment.
    - **Open Items** enables you to quickly view and access files that are currently open in your SAS Studio session. You can see that the list of open files matches the tabs.
    - **SAS Server** provides access to files and folders in your SAS environment. Keep in mind that the files and folders that you see here are not on your local machine. They're on the server where SAS is running. If you have the available permissions, you can use the **Upload files** button to load local files to the SAS server. The SAS Server also enables you to perform other tasks, such as background submit and schedule programs and basic file operations, such as copy, paste, and delete.
    - **Steps** enables you to build flows in SAS Studio. In a flow, you can build a sequence of steps, including programs, queries, and data transformations, as well as your own custom steps.
    - **Tasks** enables you to access tasks in SAS Studio. Tasks are based on SAS procedures and generate SAS code and formatted results for you.
    - **Snippets** enables you to access snippets. Snippets are lines of commonly used code or text that you can save and reuse. SAS Studio provides several predefined snippets, and you can create your own.
    - **Library Connections** provides access to your SAS libraries. You can open SAS tables and add them to your programs. You can even expand a table and view its columns.
    - **Git Repositories** enables you to access the basic Git features from within SAS Studio for maintaining and viewing a history of program edits.

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Demo 1: Working with the Compute Server

Modify a SAS®9 Program to Execute on the SAS Compute Server

6. In SAS Server, expand **Files** > **Home** > **workshop** > **Viya_Prog**. Double-click **Demo_01.sas**.
7. Demo 1 includes code that was originally written for SAS®9. When executed in SAS Viya, the program is processed on the Compute Server. Examine the program and notice that it includes traditional code to access data, create a new table, and generate summary reports.
8. What needs to change to run this program in SAS Viya? The path in the LIBNAME statement must be updated to reference the location of the data in the new environment. Highlight the path in quotation marks and then use the SAS Server to right-click the **Viya_Prog** folder and select **Insert as path**.

```
libname mydata "/home/student/workshop/Viya_Prog";
```

9. No additional code modifications are necessary. Highlight the code in Demo 1 and click **Run**.
10. This program executes on the Compute Server. It processes the data just like the SAS®9 Workspace Server and returns results for the CONTENTS and FREQ procedures, and output tables for the DATA step and the MEANS procedure.
11. Examine the log and note the execution time for each step.

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Demo 2: Working with the CAS Server

Modify a SAS Program to Execute in CAS

Start a CAS Session

12. Open **Demo_02.sas** from SAS Server. Expand **Files** > **Home** > **workshop** > **Viya_Prog**.
13. One of the benefits of working in CAS is that data is held in-memory, reducing I/O. To process data in-memory, we must first create a CAS session. Insert a line at the top of the program and add the following CAS statement to initiate a CAS session:

```
cas mySession;

libname mydata "/home/student/workshop/Viya_Prog";
```

The session encapsulates or “keeps a record” of our work and connects us to the CAS Server.

Define a Caslib

14. After establishing a session, connect data to the CAS Server using caslibs. Caslibs connect to a variety of data sources such as data in the cloud, databases, folder paths, and streaming data. The **Orders** table must be loaded into memory. First, you must define a caslib that enables CAS to read and load the data source file from disk. The caslib points to the location where your data source files are stored.
15. Before the LIBNAME statement, add the following CASLIB statement:

```
caslib mycas path="/home/student/workshop/Viya_Prog"
libref=mycas;
```

- **mycas** is the caslib name.
- PATH= defines the path to the data.
- LIBREF= maps a library reference to the caslib that can be used in DATA and PROC steps. It is recommended that the libref matches the caslib name.

16. Highlight and run the CAS and CASLIB statements. Confirm in the log that the **mycas** caslib was added. Select the Library Connections pane. Notice that **mycas** is displayed with a cloud icon, indicating the library is mapped to a caslib. **mycas** appears empty because there are no in-memory tables loaded yet.

Modify and Load a File into Memory via the DATA Step

17. The next step is to load data source files into memory. In-memory tables are copies of the original files, which remain unchanged on disk. There are multiple methods for loading data into memory. First, we start with the DATA step. This enables you to make modifications to a table before loading it to CAS. If you designate an in-memory table by specifying a caslib in the DATA statement as your output table, the step executes on the Compute Server, but the result is loaded into memory.

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18. Modify the DATA statement to write the output table to **mycas**. Highlight the DATA step and click **Run**.

```
data mycas.orders_clean;
```

19. This step processes on the Compute Server, but the result is an in-memory table available in CAS. Expand **mycas** in the Library Connections pane and confirm that **ORDERS** is listed. The lightning icon indicates in-memory tables.

Load a File into Memory via PROC CASUTIL

20. Another method to load a data source file into memory is using PROC CASUTIL. Files from a client location, a SAS dataset in a traditional SAS library, or a data source file in a caslib can be loaded into memory using PROC CASUTIL. In this demo, we load a SASHDAT file into memory. The SASHDAT file format is optimized for loading data in CAS to the available worker nodes. It supports parallel loading, meaning that data can be loaded simultaneously to the multiple workers, which can significantly reduce load time.

21. In the Snippets pane, expand **Viya Foundation** > **Cloud Analytic Services** and double-click **Load Data to caslib**. There are multiple options to load data into memory. This sample code includes three scenarios for loading data and template code for each one. Highlight the last PROC CASUTIL step and press CTRL+C to copy the code.

22. Return to the **Demo_02.sas** tab and press CTRL+V to paste the snippet before the DATA step. Modify the values in quotation marks to read the **orders.sashdat** file from the **mycas** caslib and then create an in-memory table in the **mycas** caslib named **orders**. Add the REPLACE option to replace the in-memory table if it exists.

```
proc casutil;

   load casdata="orders.sashdat" incaslib="mycas"
        outcaslib="mycas" casout="orders" replace;

run;
```

23. Highlight and run the PROC CASUTIL step. Examine the log and confirm that the **Orders** table is available in the **mycas** caslib.

24. Expand **mycas** in the Library Connections pane and confirm that **ORDERS** is listed.

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Process In-Memory Tables in CAS

Run a DATA Step in CAS

25. If both the input table on the SET statement and the output table on the DATA statement are in-memory tables, the entire step can process in CAS. For complex DATA steps and large tables, this can result in significantly faster execution.

26. Change the input table to **mycas.orders** and then rerun this DATA step. The log indicates that the step ran in Cloud Analytic Services.

```
data mycas.orders_clean;
    set mycas.orders;
    ...
```

Run Procedures in CAS

27. Many SAS procedures have corresponding CAS-enabled procedures.
28. In **Demo_02.sas**, make the following changes:

    a. Change **FREQ** to **FREQTAB**.

    b. Change **MEANS** to **MDSUMMARY**.

    c. Change **Mydata** to **Mycas** to reference in-memory tables in the CONTENTS, FREQTAB, and MDSUMMARY procedures.

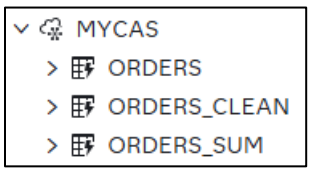
```
proc contents data=mycas.orders;
run;

proc freqtab data=mycas.orders;
    tables Country OrderType;
run;

proc mdsummary data=mycas.orders;
    var RetailPrice;
    output out=mycas.orders_sum;
run;
```

29. Highlight and run the three procedures and TITLE statements. The results look the same compared to the previous program. However, the log indicates that PROC FREQTAB and PROC MDSUMMARY executed in CAS. Behind the scenes, these steps are converted to CAS actions and then executed in CAS.

30. Select the Library Connections pane and expand **mycas**. Confirm that **ORDERS**, **ORDERS_CLEAN**, and **ORDERS_SUM** are listed. These tables are defined as session scope, which means that they are available only in SAS Studio and in the current CAS session.



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Demo 3: Understanding Caslib and Table Scope

Global Scope Caslibs and Tables

31. Open **Demo_03.sas** from SAS Server. Expand **Files** > **Home** > **workshop** > **Viya_Prog**.
32. Caslibs and tables promoted to global scope persist in memory when the CAS session ends and are available in other SAS Viya applications.
33. To create a global scope caslib, add the GLOBAL option in the CASLIB statement.

```
caslib myglbl path="/home/student/workshop/Viya_Prog/Global"
libref=myglbl global;
```

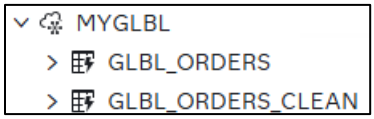
34. To promote the table using PROC CASUTIL, add the PROMOTE option in the LOAD statement.

```
proc casutil;
   load casdata="orders.sas7bdat" incaslib="myglbl"
    outcaslib="myglbl" casout="glbl_orders" promote;
quit;
```

Or you can also promote a table in the DATA step with the PROMOTE=YES data set option.

```
data myglbl.orders_clean_glbl(promote=yes);
    set myglbl.glbl_orders;
    ...
```

35. Run the program. Select the Library Connections pane and expand **myglbl**. Confirm that **GLBL_ORDERS**, and **GLBL_ORDERS_CLEAN** are listed.



36. Click the **applications menu** icon in the upper left corner. Select **Explore and Visualize** to open Visual Analytics. In the upper right corner, select **New report**. On the **Data** tab, select **Add data**. Confirm that **GLBL_ORDERS**, and **GLBL_ORDERS_CLEAN** are available.

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Demo 4: Benchmarking Traditional SAS Code, CAS-Enabled Steps, and CASL Code

37. Open **Demo_04.sas** from SAS Server. Expand **Files** > **Home** > **workshop** > **Viya_Prog**.
38. CAS-enabled procedures and DATA steps are converted to CAS actions behind the scenes and submitted to CAS for execution. You can write CASL code in PROC CAS to submit your own CAS actions directly. This program includes the complete code for the Compute Server steps, the CAS-enabled steps, and CASL. It also includes total timing information in the log for each scenario.
39. Click **Run** to execute the entire program. Examine the log and compare run times for each scenario. Your actual times will vary but notice that the CAS-enabled steps and CASL significantly reduce execution time compared to the Compute server program.

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HOW Slides

What is SAS Viya?
SAS Viya is a modern, cloud-enabled analytics platform built for high-speed processing and scalability. With just a few modifications to your existing programs, you can unlock the full power of the platform.
In this session, we will:

- Explore how to leverage your existing SAS skills in Viya

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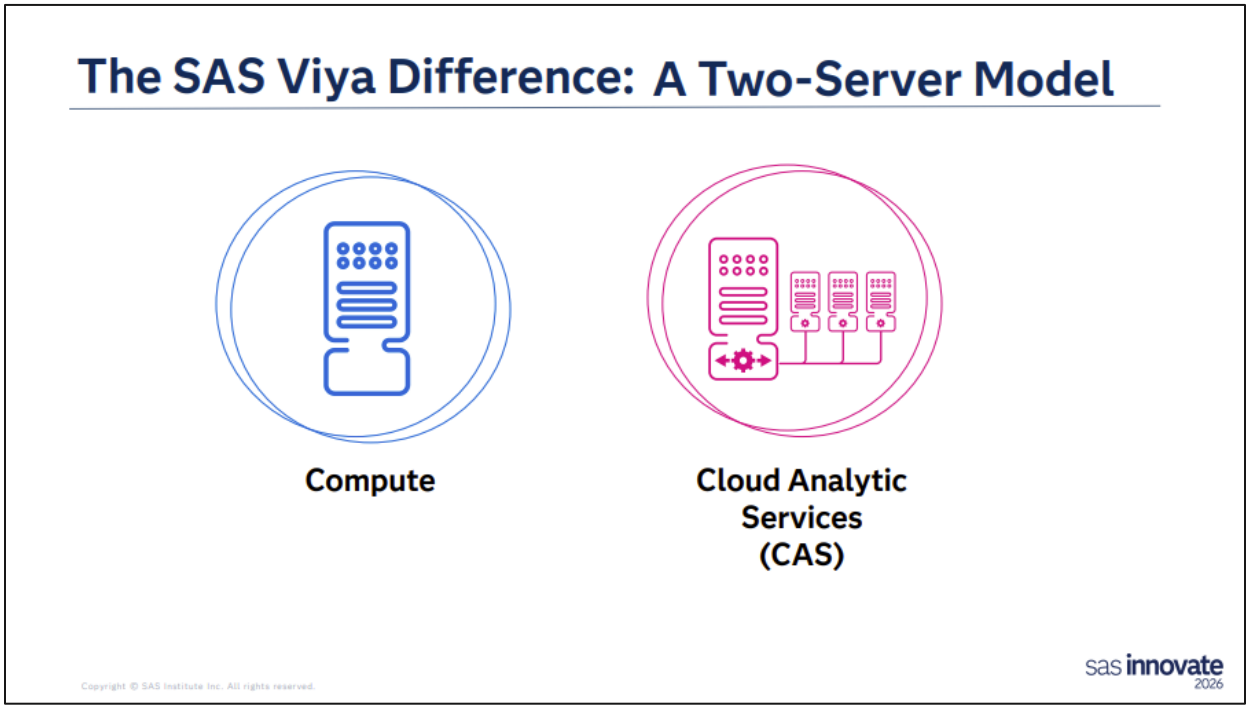
- Understand the roles of the Compute and CAS servers
- Learn about caslibs and working with in-memory data

Let's get started!



In SAS Viya, there are two primary servers, providing options to programmers depending on the processing demands of their programs.

The two servers are:

- SAS Compute Server
- CAS (Cloud Analytic Services) Server

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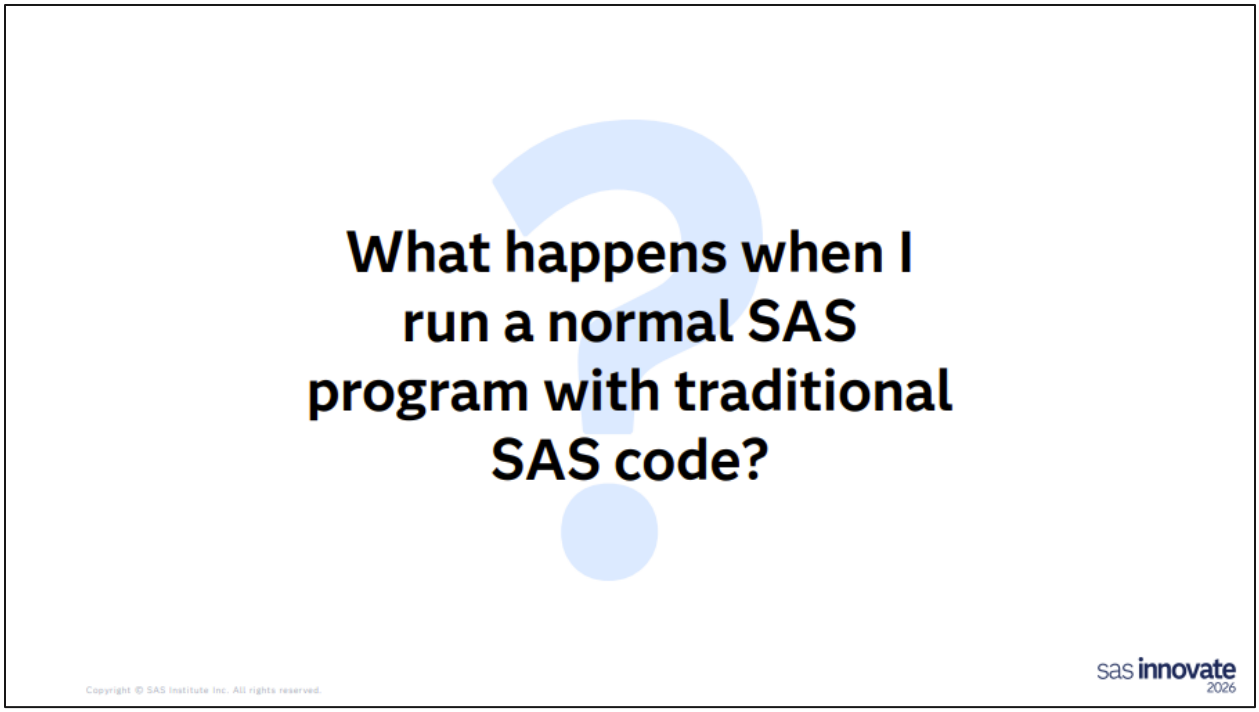
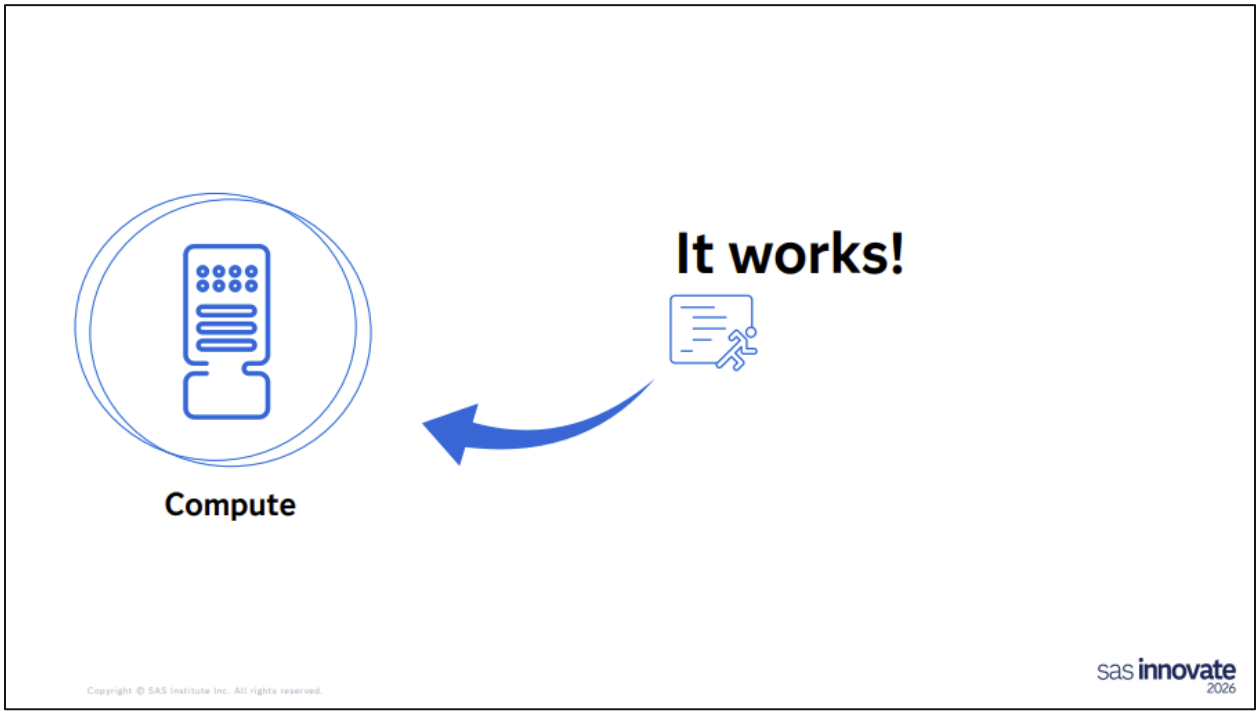


What happens when I run a normal SAS program with traditional SAS code?



It works! You can run your existing SAS programs with normal SAS code in SAS Viya and it will execute on the Compute Server. It’s a great option for smaller datasets or if you want to transition existing code into SAS Viya with minimal modifications. The Compute Server works similarly to the traditional SAS 9 Server, which could have been referred to as PC SAS or the SAS Workspace Server. One modification you will need to make is updating paths to data. Let's take a look at Demo 1.

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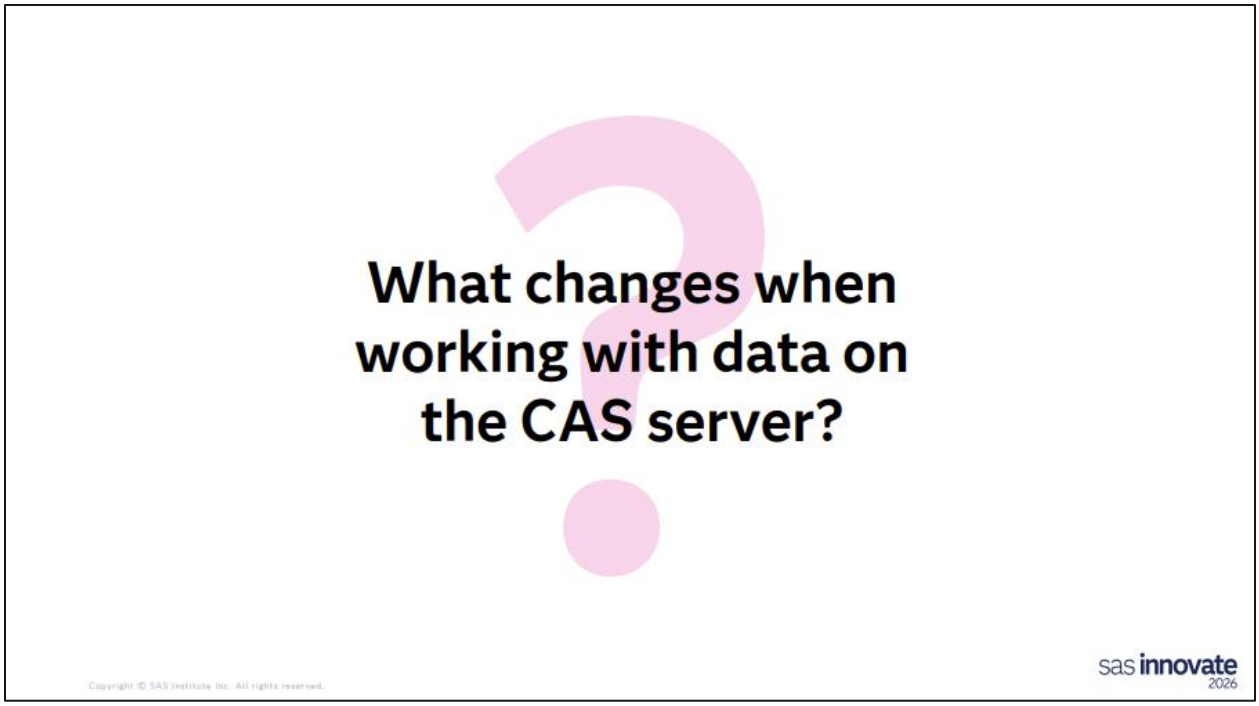
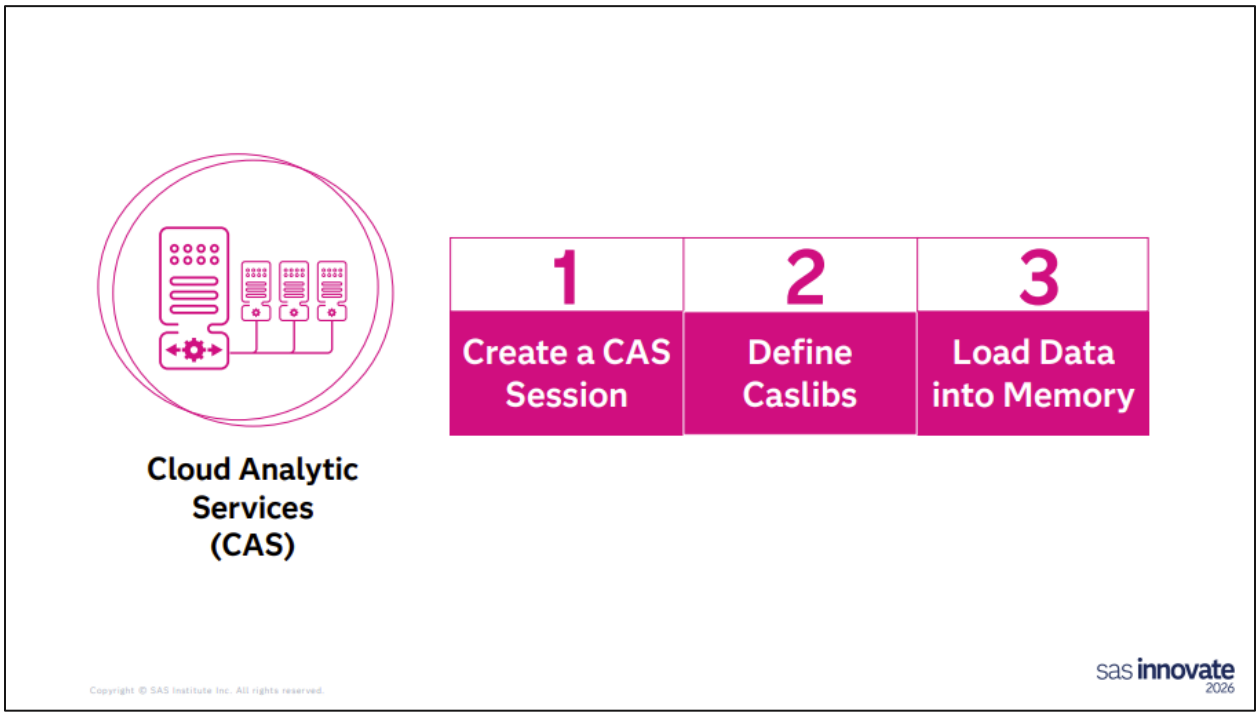


What changes when working with data on the CAS server?



One of the benefits of working in CAS is that data is held in-memory, reducing I/O. To process data in-memory, we must create a CAS session, define caslibs, and load data into memory.

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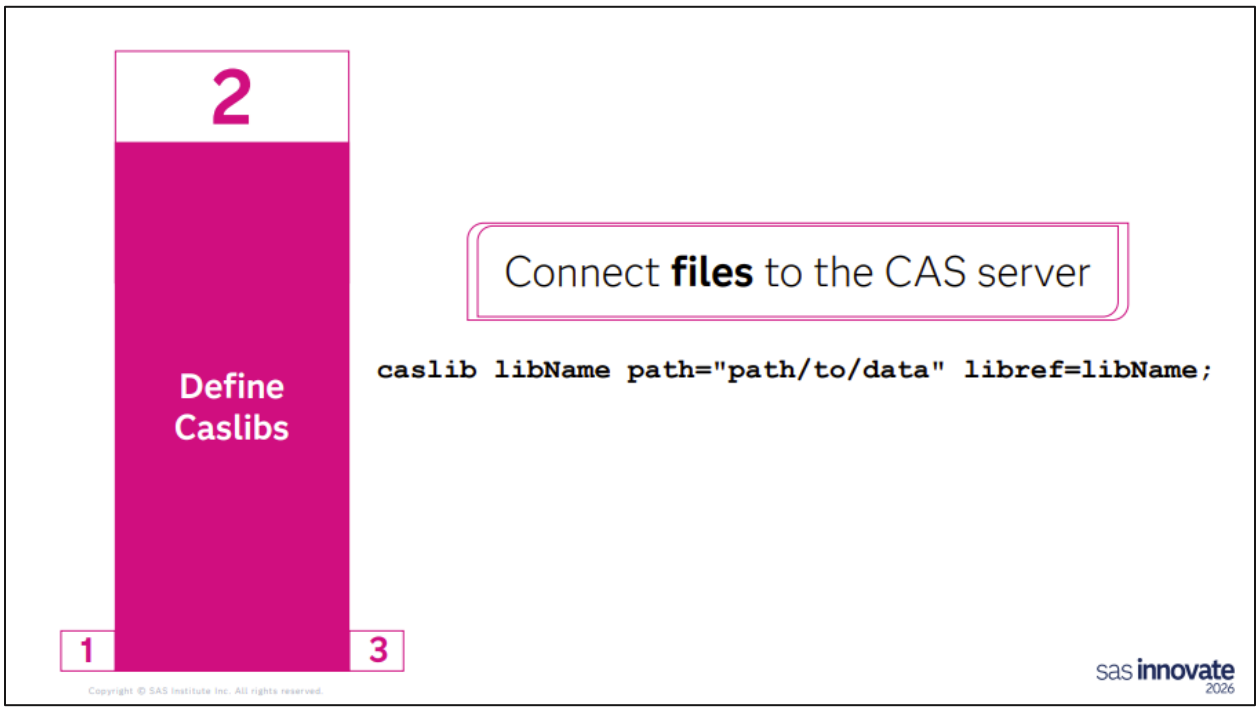
The session encapsulates or “keeps a record” of our work and connects us to the CAS Server. Use a CAS statement to create a session.



After creating a CAS session, data is connected to the CAS server using caslibs. Traditional SAS libraries connect data to the Compute server, just like caslibs connect data to the CAS server. Caslibs connect to a variety of data sources such as data in the cloud, databases, folder paths, and streaming data.

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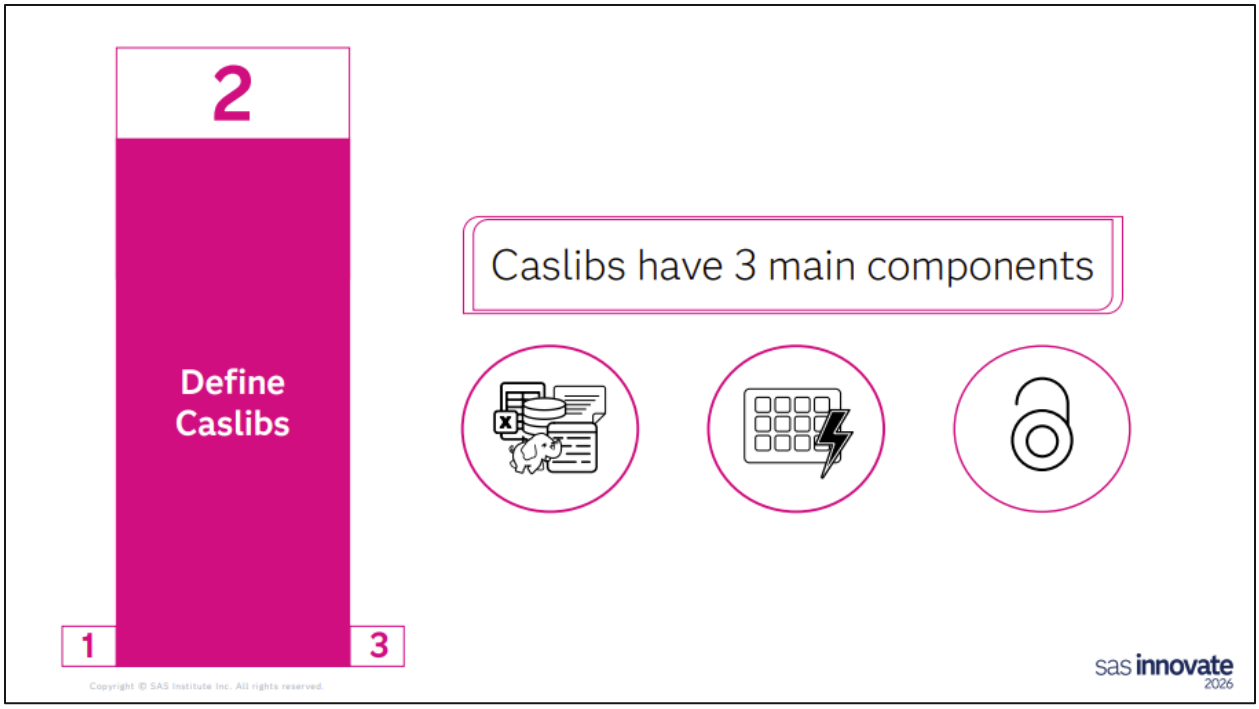
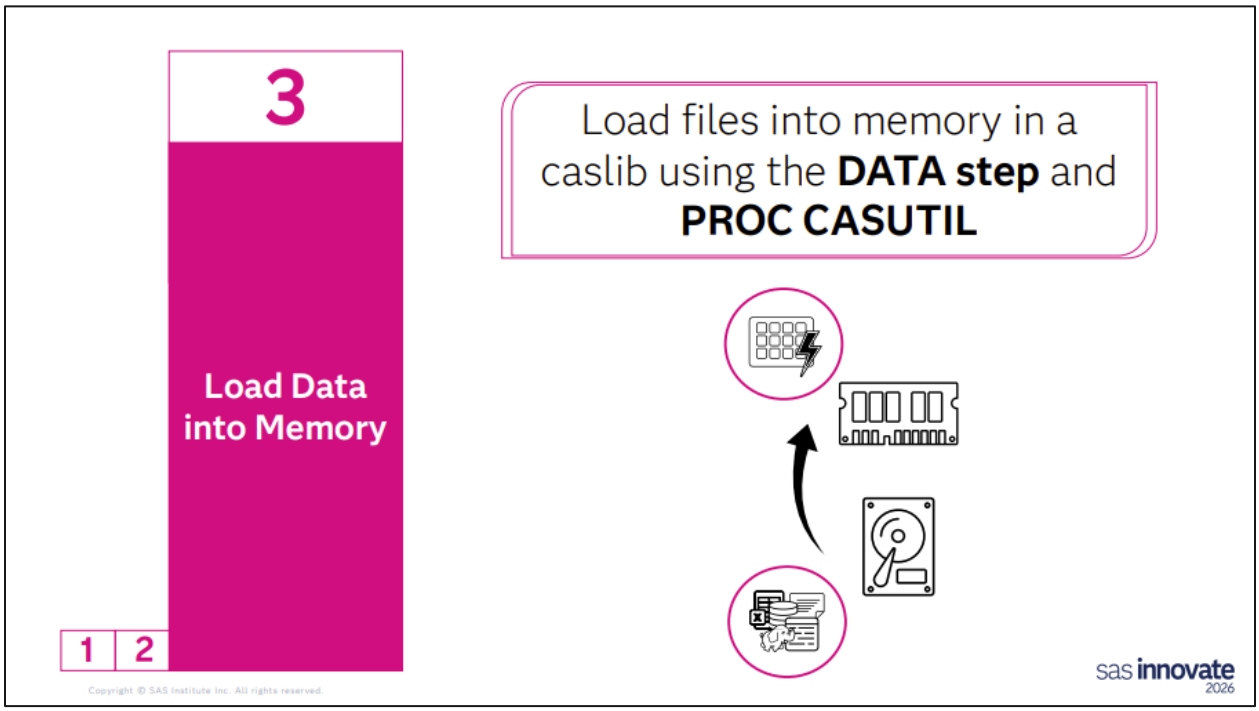


Traditional SAS libraries have one main component, the connection to the data source file. Caslibs have three main components:

1. The connection to the data source files.
2. The in-memory portion for data that has been loaded into memory.
3. Access controls to define permissions to a specific caslib.



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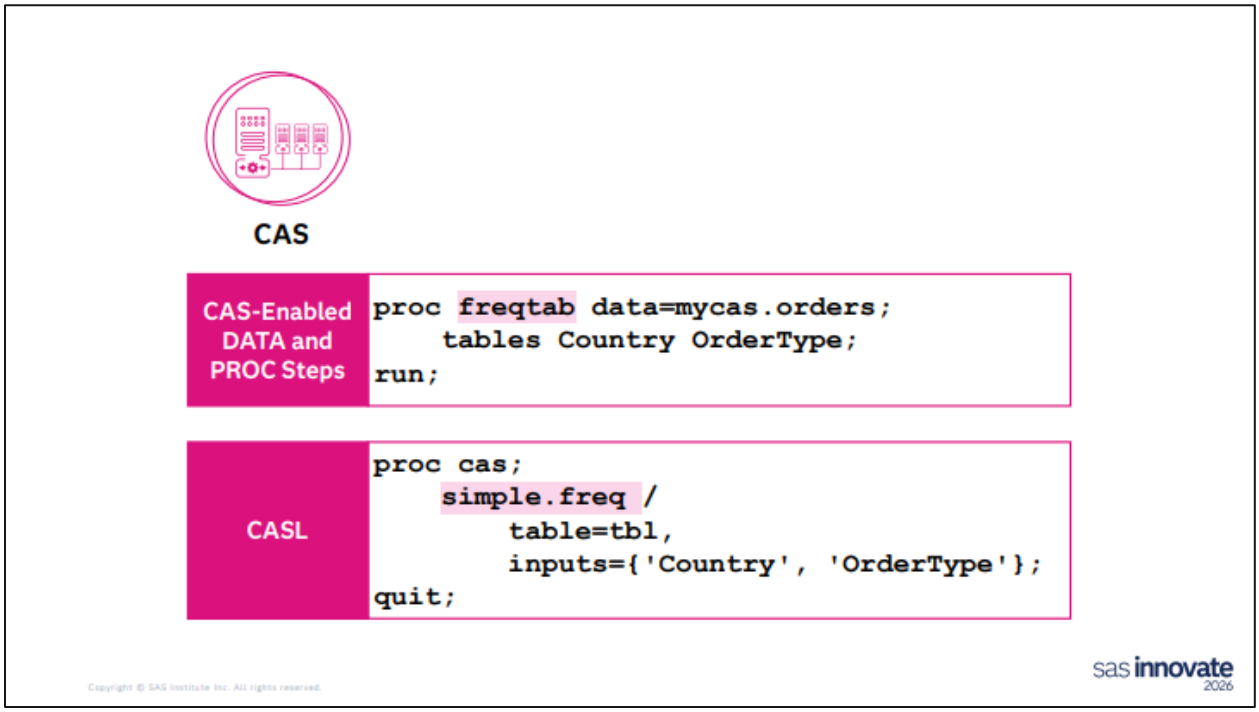
To work with data on the CAS server, you must load data source files to in-memory tables. This can be done a few ways. We'll see how to do this directly in the DATA step and introduce the CASUTIL procedure.



When working with the CAS server, there are CAS-enabled DATA and PROC steps. When code executes on the CAS server, it's turned into CAS actions behind the scenes. This is the native language of the CAS server, (CASL). We will briefly see an example of CASL and benchmark the time it takes to run in comparison to CAS-enabled steps and traditional SAS code on the Compute server. Let's jump back into a few demos.

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